SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CSA37 – SOFTWARE TESTING

EXPERIMENT 11

Case Study: Railway Reservation System – Positive Test Scenarios and Test Cases

AIM:

To identify and prepare positive test scenarios and test cases for the Railway Reservation System to verify that all major functionalities, such as train searching, seat availability checking, passenger detail entry, ticket booking, successful payment, PNR status checking, ticket downloading, and cancellation, work correctly with valid inputs and meet the specified functional requirements.

1. Problem Description

A Railway Reservation System allows passengers to search trains, check availability, enter passenger details, book tickets, pay and check or cancel bookings. Testing verifies correct operation with valid inputs.

2. Functional Requirements

1. Users must search using valid source, destination and journey date.
2. Train schedules and seat availability must be accurate.
3. Passenger details must satisfy required formats and limits.
4. A confirmed booking must have a unique PNR.
5. Ticket confirmation depends on successful payment and booking response.
6. Users may view, retrieve and cancel only authorized bookings.
7. Invalid searches and booking attempts must show clear messages.

3. Positive Test Cases

Positive testing verifies expected behavior using valid inputs.

Test Case ID	Test Scenario	Test Input / Action	Expected Result	Status
PT01	Search trains	Enter valid source, destination and journey date	Matching trains displayed	Pass
PT02	Check seat availability	Select a train and class	Availability displayed correctly	Pass
PT03	Enter passenger details	Provide valid name, age and gender	Passenger details accepted	Pass
PT04	Book available seat	Select available train/class and valid passenger details	Booking completed and PNR generated	Pass

PT05	Make successful payment	Use valid payment details in test environment	Payment succeeds and ticket is issued	Pass
PT06	Download ticket	Open confirmed booking	Ticket details downloaded/displayed	Pass
PT07	Check PNR status	Enter a valid PNR	Current booking status displayed	Pass
PT08	Cancel confirmed ticket	Select eligible ticket and confirm cancellation	Cancellation recorded and refund rules displayed	Pass

Conclusion

The positive test cases help verify that the Railway Reservation System meets its requirements and responds appropriately to the tested conditions.